Matthieu Chaldebas

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PROFILE

PhD Computational Biologist and Machine Learning Scientist with 5+ years of experience engineering predictive models for genome-wide variant prioritization and target discovery. Creator of 5ULTRA, an open-source ML framework validated on population-scale multi-omic data (490K+ UK Biobank whole genomes, proteomics). Proven track record of translating high-dimensional genomic data into mechanistic disease insights, with 11 publications including Nature, PNAS, and first-author in AJHG.

Core Competencies: Machine Learning & Predictive Modeling | Population-Scale Genomics (WGS) | Multi-Omics Integration | Target Identification | High-Performance Computing (HPC) | Statistical Genetics

EXPERIENCE

PhD Researcher & Bioinformatics Analyst — 2021 – Present

St. Giles Laboratory of Human Genetics of Infectious Diseases, The Rockefeller University, New York, NY

- **Engineered and deployed 5ULTRA**, a novel Python-based ML framework combining SpliceAI, uORF databases, and a Random Forest classifier to prioritize 5'UTR variants genome-wide (Published 1st author in AJHG, 2026).
- **Improved predictive accuracy** over the industry standard (CADD v1.7) by achieving an AUC of 0.82 vs. 0.75 on independent ClinVar 5'UTR variants benchmarks.
- **Pioneered rare 5'UTR variant burden testing at population scale**, analyzing 408,423 UK Biobank genomes across 59 quantitative traits.
- **Discovered 12 novel gene-phenotype associations** (including NELFCD for platelet count) entirely missed by standard models, yielding a 39% power increase in Target ID discovery over the CADD framework.
- **Demonstrated translational dosage rheostats** in human genetics, proving that 5'UTR hypomorphic variation yields burden effects quantitatively comparable to exome protein-truncating variants (PTVs).
- **Validated in-silico predictions** using large-scale Olink proteomics (n = 46,362) and MPRA readouts, proving identified cis-pQTL variants carry effect sizes >5x larger than baseline.
- **Orchestrated high-throughput WES/WGS pipelines** of 26,000+ patient samples, directly supporting computational analyses for 10 high-impact publications (Nature, PNAS, Immunity).

Data Scientist Intern — 6 months, 2021

BioMérieux, Lyon, France

- **Developed supervised Lasso ML models** for antibiotic resistance prediction, extending cross-validation techniques to successfully account for complex phylogenetic structures.
- **Implemented novelty detection algorithms** to clean and standardize heterogeneous clinical datasets, improving the reliability and accuracy of downstream model predictions.

Data Engineer Intern — 4 months, 2019

CEA Paris-Saclay, France

- Managed diverse biological datasets, performed phylogenetic analyses, and produced 3D protein structural models.

EDUCATION

PhD Candidate, Bioinformatics — 2022 – 2026 (Expected)

Université Paris Cité - Research at The Rockefeller University, St. Giles Lab. Advisors: Drs. Casanova, Cobat & Zhang

MS, Biotechnology Engineering — 2016 – 2021

Sup'Biotech, Paris, France

Exchange Semester: Immunology & Human Disease 2018
UC San Diego, La Jolla, CA

SELECTED PUBLICATIONS (11 TOTAL)

Chaldebas M, Ponsin K, Bohlen J, et al. (2026). **Genome-wide detection of human 5'UTR variants that impact protein translation.** *The American Journal of Human Genetics* 113(4), 809–827.

Chaldebas M, Casanova JL, Zhang P, et al. (in prep). **Mechanistic 5'UTR variant scoring expands rare-variant discovery in the UK Biobank.** *Manuscript in preparation*

Zhang P, Chaldebas M, Ogishi M, et al. (2023). **Genome-wide detection of human intronic AG-gain variants.** *PNAS* 120(46), e2314225120.

Arias AA, Neehus AL, et al. [incl. Chaldebas M] (2024). **Tuberculosis in otherwise healthy adults with inherited TNF deficiency.** *Nature* 633(8029), 417–425.

Ogishi M, Kitaoka K, et al. [incl. Chaldebas M] (2024). **Impaired development of memory B cells in humans deficient in PD-1 signaling.** *Immunity* 57(12), 2790–2807.

Chan YH, Liu Z, et al. [incl. Chaldebas M] (2024). **Human TMEFF1 is a restriction factor for herpes simplex virus in the brain.** *Nature* 632(8024), 390–400.

Matuozzo D, Talouarn E, et al. [incl. Chaldebas M] (2023). **Rare predicted loss-of-function variants of type I IFN immunity genes are associated with life-threatening COVID-19.** Genome medicine 15(1), 22.

TECHNICAL SKILLS

Programming: Python (primary), R, Bash

ML & Modeling: Scikit-learn, Random Forest, Lasso, SMOTE, feature engineering; AlphaGenome integration

Genomics & Statistics: Rare-variant burden testing (REGENIE), WGS/WES, scRNA-seq, bulk RNA-seq, pQTL/proteomics (Olink), GWAS, 5'UTR biology, SpliceAI, gnomAD, UK Biobank, ClinVar, COSMIC

Infrastructure: Snakemake, Docker, Git, HPC/SLURM, DNAnexus, UKB RAP

Languages: French (native), English (fluent — TOEIC 975/990)

VISA & ELIGIBILITY

J-1 visa (no 212e bar); 3 years H-1B eligibility remaining.